USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: CR_DONE Status during EQ_DONE Phase
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
DP 1.4a spec had an ECN which demands the DPTX/LTTPR DFP not to check CR_DONE during EQ Phase, but only at the end of it. USB4 spec aligns to the change in the DisplayPort spec, mandating that the DP OUT Adapter acts upon CR_DONE failure at the end of the EQ Phase.

Benefits as a result of the changes:
Align USB4 Spec to DisplayPort spec.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
None

Actual Change

(a). Section 10.4.10.2.2 DP OUT Adapter Requirements

10.4.10.2.2 DP OUT Adapter Requirements

A DP OUT Adapter receiving a SET_CONFIG Packet of type 8B10B_SET_LINK, with *LC* field other than 0h shall:

- Initiate link training with the target Link Rate and Lane Count received from the 8B10B_SET_LINK Packet.
- Link training proceeds according to the DisplayPort Specification except that a DP OUT Adapter that concludes that it needs to either reduce the Link Rate or Lane Count shall treat it as link training failure and shall not reduce the Link Rate or Lane Count.

A DP OUT Adapter which finishes the Clock Recovery Sequence (as defined in the DisplayPort Specification) shall send a SET_CONFIG Packet of type 8b10b STATUS_CR_DONE, reflecting the LANEx_CR_DONE statuses of the active lanes. The *Phase* field shall be set to 0b.

~~A DP OUT Adapter in~~ If at the end of the EQ phase, link training is not successful and a DP OUT Adapter ~~which~~ detects that the DP receiver has lost Clock Recovery on one or more of the active lanes, it ~~shall conclude that link training has failed and~~ shall send a SET_CONFIG Packet of type 8b10b STATUS_CR_DONE, reflecting the new LANEx_CR_DONE statuses of the active lanes. The *Phase* field shall be set to 1b.

If link training fails for a reason other than lost Clock Recovery, a DP OUT Adapter shall send a SET_CONFIG Packet of type STATUS_TRAINING_FAIL. The MSG Data shall be set as follows:

- LANEx_CHANNEL_EQ_DONE is equal to the value of the last read from *LANEx_CHANNEL_EQ_DONE* field in DPRX.
- LANEx_SYMBOL_LOCKED is equal to the value of the last read from *LANEx_SYMBOL_LOCKED* field in DPRX.

If link training finishes successfully, a DP OUT Adapter shall send a SET_CONFIG Packet of type 8B10B_SET_LINK, with the same *LC* and *LR* fields it received from the DP IN Adapter when link training was initiated. The DP OUT Adapter may then either transition to transmitting IDLE pattern (including SR) for both MST and SST DP Links or keep transmitting the same TPS it is currently transmitting.